

# Complex Analysis Notes: Toward Euler's Formula

## Contents

|          |                                                      |           |
|----------|------------------------------------------------------|-----------|
| <b>1</b> | <b>Complex Numbers</b>                               | <b>1</b>  |
| <b>2</b> | <b>Algebraic Operations</b>                          | <b>2</b>  |
| <b>3</b> | <b>Conjugates and Modulus</b>                        | <b>3</b>  |
| <b>4</b> | <b>Reciprocals and Division</b>                      | <b>4</b>  |
| <b>5</b> | <b>Complex Sequences</b>                             | <b>5</b>  |
| <b>6</b> | <b>Complex Series</b>                                | <b>8</b>  |
| <b>7</b> | <b>The Complex Exponential and Euler's Formula</b>   | <b>11</b> |
| <b>8</b> | <b>Argument, Polar Form, and De Moivre's Theorem</b> | <b>15</b> |

## 1 Complex Numbers

**Definition 1.1** (The complex numbers). *Define the complex numbers by*

$$\mathbb{C} = \mathbb{R}^2 \text{ with the usual vector addition and real scalar multiplication.}$$

*Equip this real vector space with multiplication*

$$(a, b)(c, d) = (ac - bd, ad + bc), \quad 1 = (1, 0), \quad i = (0, 1).$$

*With this multiplication,*

$$i^2 = (0, 1)(0, 1) = (-1, 0).$$

*After identifying  $a \in \mathbb{R}$  with  $(a, 0) \in \mathbb{C}$ , this says  $i^2 = -1$ . Every element  $(a, b) \in \mathbb{R}^2$  can be written as*

$$(a, b) = a(1, 0) + b(0, 1) = a + bi.$$

**Definition 1.2** (Real and imaginary parts). *If  $z = a + bi \in \mathbb{C}$ , define*

$$\operatorname{Re}(z) := a, \quad \operatorname{Im}(z) := b.$$

*Two complex numbers are equal precisely when their real and imaginary parts are equal:*

$$a + bi = c + di \iff a = c \text{ and } b = d.$$

## 2 Algebraic Operations

**Definition 2.1** (Complex addition). For  $z = a + bi$  and  $w = c + di$ , define

$$z + w := (a + c) + (b + d)i.$$

**Definition 2.2** (Complex multiplication in  $a + bi$  notation). For  $z = a + bi$  and  $w = c + di$ , define

$$zw := (ac - bd) + (ad + bc)i.$$

**Proposition 2.1** (Distributive law). For all  $z, w, u \in \mathbb{C}$ ,

$$z(w + u) = zw + zu.$$

*Proof.* Let  $z = a + bi$ ,  $w = c + di$ , and  $u = e + fi$ . Then

$$w + u = (c + e) + (d + f)i.$$

Hence

$$\begin{aligned} z(w + u) &= (a + bi)((c + e) + (d + f)i) \\ &= (a(c + e) - b(d + f)) + (a(d + f) + b(c + e))i \\ &= (ac - bd) + (ad + bc)i + (ae - bf) + (af + be)i \\ &= zw + zu. \end{aligned}$$

□

**Proposition 2.2** ( $\mathbb{C}$  is a field). With the addition and multiplication defined above,  $\mathbb{C}$  is a field.

*Proof.* The additive identity is  $0 = 0 + 0i$ , and the multiplicative identity is  $1 = 1 + 0i$ . Associativity, commutativity, and distributivity follow by direct expansion from the corresponding real-number laws. If  $z = a + bi$ , then the additive inverse is

$$-z := (-a) + (-b)i.$$

If  $z \neq 0$ , then  $a^2 + b^2 > 0$  and the multiplicative inverse is

$$z^{-1} := \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i.$$

A direct multiplication gives  $zz^{-1} = 1$ .

□

**Definition 2.3** (Complex subtraction). For  $z, w \in \mathbb{C}$ , define

$$z - w := z + (-w).$$

If  $z = a + bi$  and  $w = c + di$ , then

$$z - w = (a - c) + (b - d)i.$$

### 3 Conjugates and Modulus

**Definition 3.1** (Complex conjugate). *If  $z = a + bi$ , define the complex conjugate of  $z$  by*

$$\bar{z} := a - bi.$$

**Proposition 3.1** (Basic conjugate identities). *For all  $z, w \in \mathbb{C}$ ,*

$$\overline{z + w} = \bar{z} + \bar{w}, \quad \overline{zw} = \bar{z}\bar{w}, \quad \overline{\bar{z}} = z.$$

*Proof.* Each identity follows by writing  $z = a + bi$  and  $w = c + di$  and comparing real and imaginary parts.  $\square$

**Proposition 3.2** (Real part from conjugation). *For every  $z \in \mathbb{C}$ ,*

$$z + \bar{z} = 2 \operatorname{Re}(z).$$

*Proof.* If  $z = a + bi$ , then

$$z + \bar{z} = (a + bi) + (a - bi) = 2a = 2 \operatorname{Re}(z).$$

$\square$

**Definition 3.2** (Complex modulus). *If  $z = a + bi$ , define the modulus of  $z$  by*

$$|z| := \sqrt{a^2 + b^2}.$$

**Proposition 3.3** (Modulus squared). *For every  $z \in \mathbb{C}$ ,*

$$|z|^2 = z\bar{z}.$$

*Proof.* If  $z = a + bi$ , then

$$z\bar{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2.$$

$\square$

**Proposition 3.4** (Product rule for modulus). *For all  $z, w \in \mathbb{C}$ ,*

$$|zw| = |z| |w|.$$

*Proof.* Using the conjugate product identity,

$$|zw|^2 = zw \overline{zw} = zw \bar{z}\bar{w} = (z\bar{z})(w\bar{w}) = |z|^2 |w|^2.$$

Both sides are nonnegative, so taking square roots gives  $|zw| = |z| |w|$ .  $\square$

**Proposition 3.5** (Powers and modulus). *For every  $z \in \mathbb{C}$  and  $n \in \mathbb{N}$ ,*

$$|z^n| = |z|^n.$$

*Proof.* This follows by induction from the product rule:

$$|z^{n+1}| = |z^n z| = |z^n| |z| = |z|^{n+1}.$$

$\square$

**Proposition 3.6** (Real and imaginary parts are bounded by the modulus). *If  $z \in \mathbb{C}$ , then*

$$|\operatorname{Re}(z)| \leq |z|, \quad |\operatorname{Im}(z)| \leq |z|.$$

*Proof.* If  $z = a + bi$ , then  $a^2 \leq a^2 + b^2$  and  $b^2 \leq a^2 + b^2$ . Taking square roots gives the result.  $\square$

**Proposition 3.7** (Triangle inequality). *For all  $z, w \in \mathbb{C}$ ,*

$$|z + w| \leq |z| + |w|.$$

*Proof.* Since  $\operatorname{Re}(\alpha) \leq |\alpha|$  for every  $\alpha \in \mathbb{C}$ ,

$$\begin{aligned} |z + w|^2 &= (z + w)(\bar{z} + \bar{w}) \\ &= |z|^2 + |w|^2 + z\bar{w} + w\bar{z} \\ &= |z|^2 + |w|^2 + 2\operatorname{Re}(z\bar{w}) \\ &\leq |z|^2 + |w|^2 + 2|z\bar{w}| \\ &= |z|^2 + |w|^2 + 2|z||w| \\ &= (|z| + |w|)^2. \end{aligned}$$

Taking square roots gives the triangle inequality.  $\square$

**Proposition 3.8** (Reverse triangle inequality). *For all  $z, w \in \mathbb{C}$ ,*

$$||z| - |w|| \leq |z - w|.$$

*Proof.* By the triangle inequality,

$$|z| = |(z - w) + w| \leq |z - w| + |w|,$$

so  $|z| - |w| \leq |z - w|$ . Interchanging  $z$  and  $w$  gives

$$|w| - |z| \leq |w - z| = |z - w|.$$

Therefore  $||z| - |w|| \leq |z - w|$ .  $\square$

## 4 Reciprocals and Division

**Definition 4.1** (Complex reciprocal). *If  $z \in \mathbb{C}$  and  $z \neq 0$ , define*

$$\frac{1}{z} := \frac{\bar{z}}{|z|^2}.$$

**Proposition 4.1** (The reciprocal is the multiplicative inverse). *If  $z \neq 0$ , then*

$$z \cdot \frac{1}{z} = 1.$$

*Proof.* Using  $|z|^2 = z\bar{z}$ ,

$$z \cdot \frac{1}{z} = z \cdot \frac{\bar{z}}{|z|^2} = \frac{z\bar{z}}{|z|^2} = 1.$$

$\square$

**Definition 4.2** (Complex division). *For  $w, z \in \mathbb{C}$  with  $z \neq 0$ , define*

$$\frac{w}{z} := w \cdot \frac{1}{z} = \frac{w\bar{z}}{|z|^2}.$$

## 5 Complex Sequences

**Definition 5.1** (Complex sequence). *A complex sequence is a function*

$$z : \mathbb{N} \rightarrow \mathbb{C}.$$

*We write  $z_n$  for  $z(n)$ . If  $z_n = a_n + ib_n$ , then  $(a_n)$  and  $(b_n)$  are real sequences, and*

$$a_n = \operatorname{Re}(z_n), \quad b_n = \operatorname{Im}(z_n).$$

**Definition 5.2** ( $\varepsilon$ - $N$  convergence in  $\mathbb{C}$ ). *Let  $(z_n)$  be a complex sequence and let  $z \in \mathbb{C}$ . We say*

$$z_n \rightarrow z$$

*if for every  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  such that*

$$n > N \implies |z_n - z| < \varepsilon.$$

**Proposition 5.1** (Convergence as distance to the limit). *Let  $(z_n)$  be a complex sequence and let  $z \in \mathbb{C}$ . Then*

$$z_n \rightarrow z \iff |z_n - z| \rightarrow 0$$

*as a real sequence.*

*Proof.* The sequence  $(|z_n - z|)$  is a real sequence because each term is a nonnegative real number.

First suppose  $z_n \rightarrow z$  in  $\mathbb{C}$ . By definition, for every  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  such that

$$n > N \implies |z_n - z| < \varepsilon.$$

This is exactly the statement that the real sequence  $|z_n - z|$  converges to 0.

Conversely, suppose  $|z_n - z| \rightarrow 0$  as a real sequence. Then for every  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  such that

$$n > N \implies ||z_n - z| - 0| < \varepsilon.$$

Since  $|z_n - z| \geq 0$ ,

$$||z_n - z| - 0| = |z_n - z|.$$

Thus

$$n > N \implies |z_n - z| < \varepsilon,$$

which is precisely  $z_n \rightarrow z$  in  $\mathbb{C}$ . □

**Theorem 5.1** (Componentwise convergence). *Let  $z_n = a_n + ib_n$  and  $z = a + ib$ . Then*

$$z_n \rightarrow z \iff a_n \rightarrow a \text{ and } b_n \rightarrow b.$$

*Equivalently,*

$$z_n \rightarrow z \iff \operatorname{Re}(z_n) \rightarrow \operatorname{Re}(z) \text{ and } \operatorname{Im}(z_n) \rightarrow \operatorname{Im}(z).$$

*Proof.* First suppose  $z_n \rightarrow z$ . Since

$$|a_n - a| = |\operatorname{Re}(z_n - z)| \leq |z_n - z|, \quad |b_n - b| = |\operatorname{Im}(z_n - z)| \leq |z_n - z|,$$

the  $\varepsilon$ - $N$  definition for  $z_n \rightarrow z$  implies  $a_n \rightarrow a$  and  $b_n \rightarrow b$ .

Conversely, suppose  $a_n \rightarrow a$  and  $b_n \rightarrow b$ . Let  $\varepsilon > 0$ . Choose  $N_1$  such that

$$n > N_1 \implies |a_n - a| < \frac{\varepsilon}{2},$$

and choose  $N_2$  such that

$$n > N_2 \implies |b_n - b| < \frac{\varepsilon}{2}.$$

Let  $N = \max\{N_1, N_2\}$ . Then for  $n > N$ ,

$$\begin{aligned} |z_n - z| &= |(a_n - a) + i(b_n - b)| \\ &= \sqrt{(a_n - a)^2 + (b_n - b)^2} \\ &\leq |a_n - a| + |b_n - b| \\ &< \varepsilon. \end{aligned}$$

Therefore  $z_n \rightarrow z$ . □

**Theorem 5.2** (Uniqueness of limits). *If  $z_n \rightarrow z$  and  $z_n \rightarrow w$ , then  $z = w$ .*

*Proof.* Write

$$z = a + ib, \quad w = c + id, \quad z_n = a_n + ib_n.$$

By componentwise convergence,

$$a_n \rightarrow a, \quad b_n \rightarrow b, \quad a_n \rightarrow c, \quad b_n \rightarrow d.$$

Real limits are unique, so  $a = c$  and  $b = d$ . Therefore  $z = w$ . □

**Theorem 5.3** (Convergent sequences are bounded). *If  $z_n \rightarrow z$ , then  $(z_n)$  is bounded: there exists  $M > 0$  such that*

$$|z_n| \leq M \quad \forall n \in \mathbb{N}.$$

*Proof.* Since  $z_n \rightarrow z$ , choose  $N \in \mathbb{N}$  such that

$$n > N \implies |z_n - z| < 1.$$

For  $n > N$ , the triangle inequality gives

$$|z_n| = |(z_n - z) + z| \leq |z_n - z| + |z| < 1 + |z|.$$

Now define

$$M := \max\{|z_1|, |z_2|, \dots, |z_N|, 1 + |z|\}.$$

Then  $|z_n| \leq M$  for every  $n \in \mathbb{N}$ . □

**Definition 5.3** (Subsequence). *Let  $(z_n)$  be a complex sequence. A subsequence is a sequence of the form*

$$z_{n_k},$$

*where*

$$n_1 < n_2 < n_3 < \dots$$

*is a strictly increasing sequence of natural numbers.*

**Proposition 5.2** (Subsequences preserve limits). *If  $z_n \rightarrow z$ , then every subsequence  $z_{n_k}$  also converges to  $z$ .*

*Proof.* Let  $\varepsilon > 0$ . Choose  $N \in \mathbb{N}$  such that

$$n > N \implies |z_n - z| < \varepsilon.$$

Since  $n_k \geq k$ , every  $k > N$  satisfies  $n_k > N$ . Thus

$$k > N \implies |z_{n_k} - z| < \varepsilon.$$

Therefore  $z_{n_k} \rightarrow z$ . □

**Definition 5.4** (Cauchy sequence in  $\mathbb{C}$ ). *A complex sequence  $(z_n)$  is Cauchy if for every  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  such that*

$$m, n > N \implies |z_n - z_m| < \varepsilon.$$

**Theorem 5.4** (Cauchy criterion in  $\mathbb{C}$ ). *A complex sequence  $(z_n)$  converges if and only if it is Cauchy.*

*Proof.* Write  $z_n = a_n + ib_n$ .

Suppose  $z_n \rightarrow z$ . Every convergent sequence is Cauchy: given  $\varepsilon > 0$ , choose  $N$  such that

$$n > N \implies |z_n - z| < \frac{\varepsilon}{2}.$$

Then for  $m, n > N$ ,

$$|z_n - z_m| \leq |z_n - z| + |z_m - z| < \varepsilon.$$

Conversely, suppose  $(z_n)$  is Cauchy. Since

$$|a_n - a_m| \leq |z_n - z_m|, \quad |b_n - b_m| \leq |z_n - z_m|,$$

the real sequences  $(a_n)$  and  $(b_n)$  are Cauchy. Because  $\mathbb{R}$  is complete, there exist  $a, b \in \mathbb{R}$  such that

$$a_n \rightarrow a, \quad b_n \rightarrow b.$$

By componentwise convergence,  $z_n \rightarrow a + ib$ . □

**Theorem 5.5** (Limit algebra). *Suppose  $z_n \rightarrow z$  and  $w_n \rightarrow w$  in  $\mathbb{C}$ , and let  $\alpha \in \mathbb{C}$ . Then*

$$z_n + w_n \rightarrow z + w, \quad z_n - w_n \rightarrow z - w, \quad \alpha z_n \rightarrow \alpha z, \quad z_n w_n \rightarrow zw.$$

*If  $w \neq 0$  and  $w_n \neq 0$  eventually, then*

$$\frac{z_n}{w_n} \rightarrow \frac{z}{w}.$$

*Proof.* For sums,

$$|(z_n + w_n) - (z + w)| \leq |z_n - z| + |w_n - w| \rightarrow 0.$$

Subtraction follows from addition and scalar multiplication by  $-1$ . For scalar multiplication,

$$|\alpha z_n - \alpha z| = |\alpha| |z_n - z| \rightarrow 0.$$

For products,

$$\begin{aligned} z_n w_n - z w &= z_n w_n - z_n w + z_n w - z w \\ &= z_n (w_n - w) + (z_n - z) w. \end{aligned}$$

Since  $(z_n)$  is convergent, it is bounded; choose  $M > 0$  such that  $|z_n| \leq M$  for all  $n$ . Then

$$|z_n w_n - z w| \leq M |w_n - w| + |w| |z_n - z| \rightarrow 0.$$

For reciprocals, assume  $w \neq 0$ . Since  $w_n \rightarrow w$ , choose  $N$  such that

$$n > N \implies |w_n - w| < \frac{|w|}{2}.$$

Then for  $n > N$ ,

$$|w_n| \geq |w| - |w_n - w| > \frac{|w|}{2}.$$

Therefore

$$\left| \frac{1}{w_n} - \frac{1}{w} \right| = \left| \frac{w - w_n}{w_n w} \right| = \frac{|w_n - w|}{|w_n| |w|} \leq \frac{2}{|w|^2} |w_n - w| \rightarrow 0.$$

Thus  $1/w_n \rightarrow 1/w$ , and the quotient rule follows from the product rule:

$$\frac{z_n}{w_n} = z_n \cdot \frac{1}{w_n} \rightarrow z \cdot \frac{1}{w}.$$

□

**Proposition 5.3** (Limit of the modulus). *If  $z_n \rightarrow z$ , then*

$$|z_n| \rightarrow |z|.$$

*Proof.* By the reverse triangle inequality,

$$||z_n| - |z|| \leq |z_n - z|.$$

Since  $|z_n - z| \rightarrow 0$ , it follows that  $|z_n| \rightarrow |z|$ . □

**Proposition 5.4** (Limit of the conjugate). *If  $z_n \rightarrow z$ , then*

$$\overline{z_n} \rightarrow \overline{z}.$$

*Proof.* Since

$$|\overline{z_n} - \overline{z}| = |\overline{z_n - z}| = |z_n - z|,$$

we have  $\overline{z_n} \rightarrow \overline{z}$ . □

## 6 Complex Series

**Definition 6.1** (Partial sum sequence). *Let  $(z_n)_{n=0}^{\infty}$  be a complex sequence. The associated partial sum sequence  $(s_N)_{N=0}^{\infty}$  is defined by*

$$s_N := \sum_{n=0}^N z_n = z_0 + z_1 + \cdots + z_N.$$



**Definition 6.2** (Infinite sum). Let  $(z_n)_{n=0}^{\infty}$  be a complex sequence with partial sums

$$s_N = \sum_{n=0}^N z_n.$$

If the partial sum sequence converges to some  $S \in \mathbb{C}$ , then the infinite series converges and we write

$$\sum_{n=0}^{\infty} z_n = S.$$

Equivalently,

$$\sum_{n=0}^{\infty} z_n = S \iff \lim_{N \rightarrow \infty} \sum_{n=0}^N z_n = S.$$

**Definition 6.3** (Absolute convergence). A complex series  $\sum_{n=0}^{\infty} z_n$  is absolutely convergent if the real series

$$\sum_{n=0}^{\infty} |z_n|$$

converges.

**Theorem 6.1** (Absolute convergence implies convergence). If  $\sum_{n=0}^{\infty} z_n$  is absolutely convergent, then  $\sum_{n=0}^{\infty} z_n$  converges.

*Proof.* Let

$$s_N := \sum_{n=0}^N z_n, \quad t_N := \sum_{n=0}^N |z_n|.$$

Since  $\sum |z_n|$  converges in  $\mathbb{R}$ , the real partial sums  $(t_N)$  are Cauchy. Let  $\varepsilon > 0$ . Choose  $N \in \mathbb{N}$  such that

$$q > p > N \implies \sum_{n=p+1}^q |z_n| = t_q - t_p < \varepsilon.$$

Then for  $q > p > N$ ,

$$|s_q - s_p| = \left| \sum_{n=p+1}^q z_n \right| \leq \sum_{n=p+1}^q |z_n| < \varepsilon.$$

Thus  $(s_N)$  is Cauchy in  $\mathbb{C}$ , so it converges by the Cauchy criterion in  $\mathbb{C}$ . Therefore  $\sum z_n$  converges.  $\square$

**Theorem 6.2** (Reordering absolutely convergent series). Suppose  $\sum_{n=0}^{\infty} z_n$  is absolutely convergent and

$$\sum_{n=0}^{\infty} z_n = S.$$

If  $\sigma : \mathbb{N}_0 \rightarrow \mathbb{N}_0$  is a bijection, then

$$\sum_{n=0}^{\infty} z_{\sigma(n)}$$

also converges absolutely, and

$$\sum_{n=0}^{\infty} z_{\sigma(n)} = S.$$

*Proof.* First, the reordered absolute series converges. For every  $N$ , the finite set  $\{\sigma(0), \dots, \sigma(N)\}$  is contained in  $\{0, \dots, M\}$  for some  $M$ , so

$$\sum_{n=0}^N |z_{\sigma(n)}| \leq \sum_{n=0}^M |z_n| \leq \sum_{n=0}^{\infty} |z_n|.$$

Thus the partial sums of  $\sum |z_{\sigma(n)}|$  are increasing and bounded, hence converge.

Now let

$$s_N := \sum_{n=0}^N z_n, \quad r_N := \sum_{n=0}^N z_{\sigma(n)}.$$

Let  $\varepsilon > 0$ . Since  $\sum |z_n|$  converges and  $s_K \rightarrow S$ , choose  $K$  such that

$$\sum_{j=K+1}^{\infty} |z_j| < \varepsilon \quad \text{and} \quad |s_K - S| < \varepsilon.$$

Choose  $N$  large enough that

$$\{0, 1, \dots, K\} \subseteq \{\sigma(0), \sigma(1), \dots, \sigma(N)\}.$$

For  $m > N$ , the difference  $r_m - s_K$  is a finite sum of terms  $z_j$  with  $j > K$ . Therefore

$$|r_m - s_K| \leq \sum_{\substack{j > K \\ j \in \{\sigma(0), \dots, \sigma(m)\}}} |z_j| \leq \sum_{j=K+1}^{\infty} |z_j| \leq \varepsilon.$$

Then for  $m > N$ ,

$$|r_m - S| \leq |r_m - s_K| + |s_K - S| < 2\varepsilon.$$

Hence  $r_m \rightarrow S$ , so the reordered series converges to the same sum.  $\square$

**Theorem 6.3** (Ratio test). *Let  $\sum_{n=0}^{\infty} z_n$  be a complex series, and suppose  $z_n \neq 0$  for all sufficiently large  $n$ . If*

$$\lim_{n \rightarrow \infty} \frac{|z_{n+1}|}{|z_n|} = L,$$

*then:*

$$L < 1 \implies \sum_{n=0}^{\infty} z_n \text{ converges absolutely,}$$

$$L > 1 \implies \sum_{n=0}^{\infty} z_n \text{ diverges,}$$

$$L = 1 \implies \text{the test is inconclusive.}$$

*Proof.* First suppose  $L < 1$ . Choose  $r$  such that

$$L < r < 1.$$

Since  $|z_{n+1}|/|z_n| \rightarrow L$ , there exists  $N \in \mathbb{N}$  such that

$$n > N \implies \frac{|z_{n+1}|}{|z_n|} < r.$$

Thus, for  $k \geq 1$ ,

$$|z_{N+k}| < r^k |z_N|.$$

Therefore

$$\sum_{k=1}^{\infty} |z_{N+k}| \leq |z_N| \sum_{k=1}^{\infty} r^k < \infty.$$

Adding finitely many initial terms does not affect convergence, so  $\sum |z_n|$  converges. Hence  $\sum z_n$  converges absolutely.

Now suppose  $L > 1$ . Choose  $r > 1$  such that

$$1 < r < L.$$

Then for all sufficiently large  $n$ ,

$$\frac{|z_{n+1}|}{|z_n|} > r,$$

so  $|z_{n+1}| > r |z_n|$  eventually. In particular,  $|z_n|$  does not converge to 0. Hence  $z_n$  does not converge to 0, so the series  $\sum z_n$  diverges.

When  $L = 1$ , the test gives no conclusion. For example,  $\sum 1/n$  diverges but  $\sum 1/n^2$  converges, and both have ratio limit 1.  $\square$

## 7 The Complex Exponential and Euler's Formula

**Theorem 7.1** (Binomial theorem over  $\mathbb{C}$ ). *For all  $z, w \in \mathbb{C}$  and  $n \in \mathbb{N}$ ,*

$$(z + w)^n = \sum_{k=0}^n \binom{n}{k} z^k w^{n-k}.$$

*Proof.* The proof is the same induction as over  $\mathbb{R}$ , using that  $\mathbb{C}$  is a commutative field. For  $n = 1$ , the statement is immediate. Assume it holds for  $n$ . Then

$$\begin{aligned} (z + w)^{n+1} &= (z + w) \sum_{k=0}^n \binom{n}{k} z^k w^{n-k} \\ &= \sum_{k=0}^n \binom{n}{k} z^{k+1} w^{n-k} + \sum_{k=0}^n \binom{n}{k} z^k w^{n+1-k}. \end{aligned}$$

Collecting equal powers and using Pascal's identity gives

$$(z + w)^{n+1} = \sum_{k=0}^{n+1} \binom{n+1}{k} z^k w^{n+1-k}.$$

$\square$

**Theorem 7.2** (Cauchy product theorem). *Suppose  $\sum_{n=0}^{\infty} a_n$  and  $\sum_{n=0}^{\infty} b_n$  are absolutely convergent complex series, and define*

$$c_n := \sum_{k=0}^n a_k b_{n-k}.$$

*Then  $\sum c_n$  converges absolutely and*

$$\sum_{n=0}^{\infty} c_n = \left( \sum_{n=0}^{\infty} a_n \right) \left( \sum_{n=0}^{\infty} b_n \right).$$

*Proof.* Let

$$A_* := \sum_{n=0}^{\infty} |a_n|, \quad B_* := \sum_{n=0}^{\infty} |b_n|.$$

For every  $N$ ,

$$\sum_{n=0}^N |c_n| \leq \sum_{n=0}^N \sum_{k=0}^n |a_k| |b_{n-k}| \leq A_* B_*.$$

Thus  $\sum c_n$  is absolutely convergent.

Let

$$A = \sum_{n=0}^{\infty} a_n, \quad B = \sum_{n=0}^{\infty} b_n,$$

and define

$$A_N := \sum_{n=0}^N a_n, \quad B_N := \sum_{n=0}^N b_n, \quad C_N := \sum_{n=0}^N c_n.$$

Then

$$A_N B_N = \sum_{\substack{0 \leq k \leq N \\ 0 \leq j \leq N}} a_k b_j, \quad C_N = \sum_{\substack{k, j \geq 0 \\ k+j \leq N}} a_k b_j.$$

Choose  $K$  so large that both tails

$$\sum_{k>K} |a_k|, \quad \sum_{j>K} |b_j|$$

are small. If  $N > 2K$ , then every pair satisfying  $0 \leq k, j \leq N$  and  $k+j > N$  has either  $k > K$  or  $j > K$ . Hence

$$|A_N B_N - C_N| \leq B_* \sum_{k>K} |a_k| + A_* \sum_{j>K} |b_j|,$$

which can be made arbitrarily small. Since  $A_N \rightarrow A$  and  $B_N \rightarrow B$ , we have  $A_N B_N \rightarrow AB$ . Therefore  $C_N \rightarrow AB$ .  $\square$

**Definition 7.1** (The complex exponential). For  $z \in \mathbb{C}$ , define

$$\exp(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

**Proposition 7.1** (The exponential series converges absolutely). For every  $z \in \mathbb{C}$ , the series

$$\sum_{n=0}^{\infty} \frac{z^n}{n!}$$

converges absolutely.

*Proof.* If  $z = 0$ , then

$$\sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + 0 + 0 + \dots$$

is absolutely convergent.

Now suppose  $z \neq 0$ . Let

$$a_n := \frac{z^n}{n!}.$$

Then  $a_n \neq 0$  for every  $n$ , and

$$\frac{|a_{n+1}|}{|a_n|} = \frac{|z^{n+1}|}{(n+1)!} \frac{n!}{|z^n|} = \frac{|z|}{n+1} \rightarrow 0.$$

By the ratio test,  $\sum a_n$  converges absolutely. □

**Theorem 7.3** (Exponential addition law). *For all  $z, w \in \mathbb{C}$ ,*

$$\exp(z + w) = \exp(z) \exp(w).$$

*Proof.* The exponential series for  $z$  and  $w$  converge absolutely, so their product is given by the Cauchy product:

$$\begin{aligned} \exp(z) \exp(w) &= \left( \sum_{n=0}^{\infty} \frac{z^n}{n!} \right) \left( \sum_{n=0}^{\infty} \frac{w^n}{n!} \right) \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{z^k}{k!} \frac{w^{n-k}}{(n-k)!}. \end{aligned}$$

Since

$$\frac{1}{k!(n-k)!} = \frac{1}{n!} \binom{n}{k},$$

we get

$$\begin{aligned} \exp(z) \exp(w) &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^n \binom{n}{k} z^k w^{n-k} \\ &= \sum_{n=0}^{\infty} \frac{(z+w)^n}{n!} \\ &= \exp(z+w), \end{aligned}$$

where the binomial theorem was used in the second-to-last line. □

**Definition 7.2** (Sine and cosine by power series). *For  $\theta \in \mathbb{R}$ , define*

$$\cos \theta := \sum_{n=0}^{\infty} (-1)^n \frac{\theta^{2n}}{(2n)!}, \quad \sin \theta := \sum_{n=0}^{\infty} (-1)^n \frac{\theta^{2n+1}}{(2n+1)!}.$$

**Theorem 7.4** (Euler's formula). *For every  $\theta \in \mathbb{R}$ ,*

$$\exp(i\theta) = \cos \theta + i \sin \theta.$$

*Proof.* By definition,

$$\exp(i\theta) = \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!}.$$

This series is absolutely convergent by the previous proposition, so its terms may be reordered and grouped without changing the sum. Separate the even and odd powers:

$$\exp(i\theta) = \sum_{n=0}^{\infty} \frac{(i\theta)^{2n}}{(2n)!} + \sum_{n=0}^{\infty} \frac{(i\theta)^{2n+1}}{(2n+1)!}.$$

Since

$$i^{2n} = (-1)^n, \quad i^{2n+1} = i(-1)^n,$$

we get

$$\begin{aligned} \exp(i\theta) &= \sum_{n=0}^{\infty} (-1)^n \frac{\theta^{2n}}{(2n)!} + i \sum_{n=0}^{\infty} (-1)^n \frac{\theta^{2n+1}}{(2n+1)!} \\ &= \cos \theta + i \sin \theta. \end{aligned}$$

□

**Theorem 7.5** (Real and imaginary parts of  $\exp z$ ). *Let  $z = a + ib$  with  $a, b \in \mathbb{R}$ . Then*

$$\exp(z) = \exp(a)(\cos b + i \sin b).$$

*Consequently,*

$$\operatorname{Re}(\exp z) = \exp(a) \cos b, \quad \operatorname{Im}(\exp z) = \exp(a) \sin b.$$

*Equivalently,*

$$\operatorname{Re}(\exp z) = \exp(\operatorname{Re} z) \cos(\operatorname{Im} z), \quad \operatorname{Im}(\exp z) = \exp(\operatorname{Re} z) \sin(\operatorname{Im} z).$$

*Proof.* Using the exponential addition law and Euler's formula,

$$\exp(z) = \exp(a + ib) = \exp(a) \exp(ib) = \exp(a)(\cos b + i \sin b).$$

Since  $\exp(a)$ ,  $\cos b$ , and  $\sin b$  are real,

$$\exp(z) = \exp(a) \cos b + i \exp(a) \sin b.$$

Reading off real and imaginary parts gives the result. □

**Theorem 7.6** (Modulus of the complex exponential). *For every  $z \in \mathbb{C}$ ,*

$$|\exp(z)| = \exp(\operatorname{Re} z).$$

*In particular, for every  $\theta \in \mathbb{R}$ ,*

$$|\exp(i\theta)| = 1.$$

*Proof.* Write  $z = a + ib$  with  $a, b \in \mathbb{R}$ . From the previous theorem,

$$\exp(z) = \exp(a)(\cos b + i \sin b).$$

Thus

$$\begin{aligned} |\exp(z)| &= |\exp(a)| |\cos b + i \sin b| \\ &= \exp(a) \sqrt{\cos^2 b + \sin^2 b} \\ &= \exp(a). \end{aligned}$$

Here we use the standard real-analysis facts that  $\exp(a) > 0$  for real  $a$  and  $\cos^2 b + \sin^2 b = 1$ . Since  $a = \operatorname{Re} z$ , this proves  $|\exp(z)| = \exp(\operatorname{Re} z)$ . If  $z = i\theta$ , then  $\operatorname{Re}(i\theta) = 0$ , so

$$|\exp(i\theta)| = \exp(0) = 1.$$

□

## 8 Argument, Polar Form, and De Moivre's Theorem

**Definition 8.1** (Argument). Let  $z \in \mathbb{C}$  with  $z \neq 0$ . An argument of  $z$  is any real number  $\theta$  such that

$$z = |z|(\cos \theta + i \sin \theta).$$

The set of all arguments of  $z$  is denoted

$$\arg(z).$$

Arguments are not unique: if  $\theta \in \arg(z)$ , then

$$\arg(z) = \{\theta + 2\pi k : k \in \mathbb{Z}\}.$$

**Definition 8.2** (Polar form). If  $z \neq 0$  and  $\theta \in \arg(z)$ , then the polar form of  $z$  is

$$z = r(\cos \theta + i \sin \theta), \quad r = |z| > 0.$$

Using Euler's formula, this can also be written as

$$z = re^{i\theta}.$$

**Proposition 8.1** (Existence of polar form). Every nonzero complex number has a polar form.

*Proof.* Let  $z = a + ib \neq 0$  and set  $r = |z| = \sqrt{a^2 + b^2}$ . Then  $r > 0$ , and

$$\left(\frac{a}{r}\right)^2 + \left(\frac{b}{r}\right)^2 = 1.$$

By the standard real parametrization of the unit circle, there exists  $\theta \in \mathbb{R}$  such that

$$\cos \theta = \frac{a}{r}, \quad \sin \theta = \frac{b}{r}.$$

Therefore

$$z = a + ib = r(\cos \theta + i \sin \theta) = re^{i\theta}.$$

□

**Proposition 8.2** (Multiplication in polar form). If

$$z = re^{i\theta}, \quad w = se^{i\phi},$$

with  $r, s > 0$ , then

$$zw = rs e^{i(\theta+\phi)}.$$

*Proof.* Using the exponential addition law,

$$zw = (re^{i\theta})(se^{i\phi}) = rs e^{i\theta} e^{i\phi} = rs e^{i(\theta+\phi)}.$$

□

**Theorem 8.1** (De Moivre's theorem). For every  $\theta \in \mathbb{R}$  and  $n \in \mathbb{N}$ ,

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

Equivalently,

$$(e^{i\theta})^n = e^{in\theta}.$$

*Proof.* By Euler's formula,

$$\cos \theta + i \sin \theta = e^{i\theta}.$$

Using the exponential addition law repeatedly,

$$(e^{i\theta})^n = \underbrace{e^{i\theta} \cdots e^{i\theta}}_{n \text{ factors}} = e^{i(\theta + \cdots + \theta)} = e^{in\theta}.$$

Applying Euler's formula again gives

$$e^{in\theta} = \cos(n\theta) + i \sin(n\theta).$$

□



## Useful Real and Imaginary Parts

| Expression              | $\operatorname{Re}(\cdot)$  | $\operatorname{Im}(\cdot)$  |
|-------------------------|-----------------------------|-----------------------------|
| $z = a + bi$            | $a$                         | $b$                         |
| $\bar{z} = a - bi$      | $a$                         | $-b$                        |
| $z + w, w = c + di$     | $a + c$                     | $b + d$                     |
| $zw, w = c + di$        | $ac - bd$                   | $ad + bc$                   |
| $z^{-1}, z \neq 0$      | $\frac{a}{a^2 + b^2}$       | $-\frac{b}{a^2 + b^2}$      |
| $\frac{w}{z}, z \neq 0$ | $\frac{ac + bd}{a^2 + b^2}$ | $\frac{ad - bc}{a^2 + b^2}$ |
| $\exp(a + bi)$          | $\exp(a) \cos b$            | $\exp(a) \sin b$            |
| $\exp(i\theta)$         | $\cos \theta$               | $\sin \theta$               |
| $re^{i\theta}, r > 0$   | $r \cos \theta$             | $r \sin \theta$             |